

Administrative markers of maternal near miss in Brazil: regional patterns, age composition and socioeconomic associations in SUS hospital admissions, 2002–2025

Tainá dos Santos Rêgo; Bruno G. Spinelli; Abner Cardoso Rodrigues Neto; Reginaldo Antônio de Oliveira Freitas Júnior

Edmond and Lily Safra International Institute of Neuroscience, Santos Dumont Institute, Macaíba, Brazil.

Abstract

Objective: To describe administrative markers of maternal near miss in obstetric admissions financed by Brazil's Unified Health System (SUS), examining regional and age patterns, agreement between diagnostic lists and socioeconomic associations.

Methods: We conducted a descriptive and ecological study of SIH-SUS admissions processed from 2002 to 2025. Eligible records were normal hospital authorizations for females aged 10–49 years with obstetric specialty or a Chapter O diagnosis and known discharge outcome. Prenatal registration was not required. Surviving admissions with a principal diagnosis in Waterstone-derived, Mantel-derived and WHO-related ICD-10 lists were identified. Frequencies per 1,000 obstetric admissions were calculated before and after direct age standardization. Regional socioeconomic associations were explored with bivariate correlations. A complementary municipal analysis in Rio Grande do Norte (RN), 2008–2012, followed the diagnostic approach of Rosendo and Roncalli.

Results: The common-month population comprised 58,376,917 admissions across 287 nationally complete processing months. Waterstone-CID, Mantel-CID and WHO-CID frequencies were 36.90, 9.29 and 21.09 per 1,000 admissions, respectively. The North had the highest pooled frequencies for all three lists, including after age standardization. Admissions at ages 35–49 had higher frequencies than those at ages 10–34. Prenatal registration was present in 16.47% of admissions; restriction to that subgroup reduced frequencies and changed regional rankings. Of 90 exploratory regional Pearson comparisons, 19 had multiplicity-adjusted $q <$

0.05; seven of these changed direction after first differencing. In RN, the complementary marker set identified 10,321 of 221,297 admissions, without a Pearson association surviving multiplicity adjustment across nine municipal comparisons.

Conclusion: Public SUS records reveal substantial regional and age variation in administrative markers. Population selection and diagnostic definition materially affect their interpretation. These findings support transparent administrative surveillance and hypothesis generation, while requiring distinct clinical and population-based evidence to explain underlying risks.

Keywords: maternal near miss; severe maternal morbidity; hospital information systems; health inequalities; Brazil; public health surveillance.

Introduction

Maternal near miss describes survival following a life-threatening complication during pregnancy, childbirth or the postpartum period. Studying survivors complements maternal mortality surveillance by identifying a larger group of severe events and opportunities to investigate care pathways [1]. The clinical frameworks proposed by Mantel, Waterstone and the World Health Organization (WHO) differ in their emphasis on conditions, organ dysfunction and management criteria [1, 2, 3]. Their application requires information that is not uniformly available in routine administrative records.

In Brazil, the Hospital Information System of SUS (SIH-SUS) provides public records of publicly financed hospital care over several decades. These data offer substantial geographic and temporal reach. Studies by Rosendo and Roncalli, Carvalho and colleagues, and Herdt and colleagues have used administrative definitions to investigate maternal morbidity and its territorial distribution [4, 5, 6]. Their contribution is to describe patterns detectable in the available information system; diagnosis codes alone do not establish the clinical thresholds required for every near-miss criterion. Hospital comparisons with medical records and comparisons between clinical definitions demonstrate why administrative identification and clinical confirmation must remain distinct [7, 8].

Two questions are particularly relevant to interpreting administrative patterns. First, regional and temporal differences may reflect changes in the age distribution of hospitalized women as well as changes in the recorded conditions. Crude frequencies do not separate those components. Second, imposing a requirement for a completed administrative field may select a subgroup

whose relationship to the broader obstetric population varies across place and time. Prenatal-registration fields warrant this consideration because the presence of a registration number is not equivalent to attendance or quality of prenatal care.

The geographic scale of analysis also matters. Municipal comparisons of socioeconomic context address a different question from correlations between annual regional averages. Rosendo and Roncalli's ecological study in RN provides a methodological reference for examining local heterogeneity [4], while population surveys contribute evidence on individual and household circumstances [9]. These forms of evidence are complementary and should not be treated as interchangeable estimates of the same association.

This study describes administrative markers of maternal near miss in obstetric admissions financed by SUS between 2002 and 2025. We compare three diagnostic lists, examine regional frequencies before and after age standardization, characterize diagnostic composition and list overlap, and assess sensitivity to requiring prenatal registration. We also explore regional socioeconomic associations and perform a complementary municipal analysis in RN inspired by Rosendo and Roncalli.

Methods

Design, sources and population

We used publicly available reduced SIH-SUS hospital files for processing months from January 2002 through December 2025, obtained from DATASUS on 16 September 2026 [10]. Files from subsequent processing years were excluded. The analytical unit was a hospital admission authorization (AIH), rather than a uniquely identified woman. Time was defined by processing competence, which may differ from the date of the clinical event.

Eligibility required a normal AIH, a valid identifier, female sex, age 10–49 expressed in years, and a recorded discharge outcome. An admission was considered obstetric when it had obstetric specialty or a Chapter O diagnosis in any available diagnostic field. A non-zero prenatal-registration number was not required. Admissions ending in death were included in the denominator but could not be near-miss identifications. The analysis therefore describes surviving admissions with markers among all eligible obstetric admissions.

When an AIH appeared more than once, we retained its latest observed processing competence within the study period, including later versions that no longer satisfied eligibility. Identical

analytical records at that competence counted once; conflicting records were excluded and quantified. Without a person-level linkage key, separate admissions of the same woman could not be identified.

Administrative definitions and complementary attributes

Principal diagnoses were matched against three ICD-10 lists derived from the Brazilian administrative applications of the Waterstone, Mantel and WHO frameworks [4, 6, 7, 8]. They contained 102, 158 and 262 distinct codes, respectively. Throughout this article, Waterstone-CID, Mantel-CID and WHO-CID denote these administrative lists, not clinical confirmation of the corresponding frameworks. Matching was exact after removal of punctuation; three-character categories were not automatically expanded to all subcategories. The complete codebook is supplied with the analytical materials.

An admission could belong to more than one list. We reported their union and the observed mutually exclusive positive combinations. Pairwise similarity was assessed with the Jaccard index, defined as the size of the intersection divided by the union; this measures overlap, not diagnostic validity. Diagnostic groups describe recorded conditions and do not establish severity thresholds for individual admissions. Recorded intensive-care use was defined by at least one ICU day; absent information remained unknown.

We evaluated three definition sensitivities: restricting the same obstetric population to records with a non-zero prenatal-registration number; identifying markers in any available diagnostic field; and using only surviving admissions as the denominator. Prenatal-field completeness was interpreted as a recording characteristic. Secondary-diagnosis availability and coding practices may change over time.

Geography, coverage and age standardization

The national analysis used state of processing/hospitalization and the five IBGE geographic regions. The expected inventory comprised 26 states and the Federal District, comprising 27 federative units, 24 years and 12 months, totaling 7,776 files. Amapá/October 2007 was not listed in the source. Pooled regional comparisons therefore used only the 287 processing months complete in all 27 federative units. Annual series used complete months within each territory; years with fewer than 12 months were identified as partial and excluded from annual plots and temporal correlations. No missing month was treated as zero or annualized.

Frequencies were expressed as identifications per 1,000 eligible obstetric admissions. These are not WHO maternal near-miss ratios per live births. For direct age standardization, we used eight five-year groups from 10–14 to 45–49 years. The standard weights were the age distribution of all national eligible admissions in the common-month pooled analysis. Each territory-year frequency was the weighted sum of its age-specific frequencies using these fixed weights. A descriptive frequency ratio compared ages 35–49 with 10–34, with a conventional log-scale 95% confidence interval (CI).

Regional socioeconomic associations

Six official IBGE series were obtained through the aggregate-data API [11]. PNAD series described mean nominal monthly income among persons aged 10 years or older, the percentage of that population with no schooling or up to five years of schooling, and the percentage of permanent private households with a sewer connection or septic tank. PNAD Contínua series described real mean monthly income among residents with income, the percentage of persons aged 25 years or older with no schooling or fewer than five years of schooling, and sewer/septic provision among households with a bathroom, toilet or latrine. The income series differed in population universe and monetary basis; nominal PNAD income was not interpreted as purchasing power. PNAD Contínua income used the 2025 price basis of the downloaded series.

These series were analyzed separately because their universes and definitions differed. Records were joined by region code and year, without interpolation. PNAD comparisons for the North began in 2004, after expansion of its rural coverage [12]. Analyses included only years with complete regional SIH coverage and an observed covariate.

For each indicator, region and list, we estimated Pearson correlations with Fisher 95% CIs, Spearman correlations, and Pearson correlations of first differences between consecutive years. At least four nonconstant paired observations were required. Benjamini–Hochberg adjustment was applied to the family of 90 Pearson tests using crude frequencies. The 90 age-standardized comparisons formed a separate sensitivity family. Conventional CIs and p-values assume independent observations and do not account for serial dependence; the analyses were exploratory. No multivariable models or causal effects were estimated.

Municipal analysis in Rio Grande do Norte

We performed a partial reconstruction of the design of Rosendo and Roncalli [4], using normal AIHs for RN residents aged 15–49 hospitalized in RN during 2008–2012. Eligibility required a

Chapter O principal diagnosis or an obstetric procedure, operationalized through SIGTAP prefixes 030310, 0310 and 0411. The original study's diagnostic-code table defined the overall Waterstone outcome, hypertension and hemorrhage, drawing on the coding approach of Sousa and colleagues [13]. Survivors contributed to the numerators; all eligible admissions contributed to denominators.

Municipal frequencies were paired with 2010 Census income per capita, illiteracy among persons aged 10 or older, and sewer/septic coverage. We estimated Pearson and Spearman correlations without outcome-based exclusions and repeated the analysis for municipalities with at least 1,000 admissions. Each set of nine Pearson comparisons received a separate Benjamini–Hochberg adjustment. This was not an exact replication: municipalities replaced the original spatial clusters, and Census covariates did not reproduce the entire Atlas/PNUD and care-coverage variable set. Spatial dependence and small counts limit conventional inference.

Reproducibility

The analysis plan, full code lists, source metadata, coverage inventory, Python scripts and aggregate results accompany the manuscript in the working repository. Automated checks reconcile monthly totals with age strata, regional totals with national totals, and positive identifications with their denominators and list intersections. A static public-facing dashboard presents the aggregate results and definitions. No individual microdata are redistributed with the dashboard.

Results

Coverage and study population

All 7,775 available files were processed, representing 288,381,354 hospital record versions before eligibility and deduplication. AP/October 2007 was absent from the source inventory. The common-month analysis comprised 58,376,917 eligible obstetric AIHs in 287 months, including 20,566 admissions ending in death. Non-zero prenatal registration was present in 9,613,424 admissions (16.47%).

Diagnostic lists and regional frequencies

Waterstone-CID identified 2,154,047 admissions (36.90 per 1,000); Mantel-CID identified 542,154 admissions (9.29 per 1,000); WHO-CID identified 1,231,310 admissions (21.09 per 1,000). Their union included 2,271,149 admissions. Table 1 presents crude and age-standardized

regional frequencies. The North had the highest pooled frequency for each list and the South the lowest; this ordering persisted after age standardization. The pooled ordering did not describe every year: by 2025, the Northeast had the highest Waterstone-CID frequency (Figure 1). Nationally, Waterstone-CID increased from 28.74 per 1,000 in 2002 to 45.65 in 2025; the corresponding standardized frequencies were 29.66 and 44.23. Mantel-CID frequencies were 11.46 and 11.01, and WHO-CID frequencies were 21.06 and 21.21, at those endpoints. These endpoint comparisons are descriptive and do not constitute tests of temporal trend.

Table 1. Crude / age-standardized frequencies per 1,000 obstetric admissions, 2002–2025.

The same 287 nationally complete months contribute to every region. W = Waterstone-CID; M = Mantel-CID. Fixed weights are the national pooled age distribution.

Region	Admissions	W	M	WHO-CID
Brazil	58,376,917	36.90 / 36.90	9.29 / 9.29	21.09 / 21.09
North	6,805,084	42.06 / 43.32	11.89 / 11.98	29.24 / 29.85
Northeast	18,872,990	40.58 / 40.85	9.25 / 9.27	20.89 / 20.96
Southeast	20,851,674	34.20 / 33.76	8.67 / 8.64	19.47 / 19.32
South	7,356,343	31.90 / 31.37	8.13 / 8.08	18.70 / 18.49
Central-West	4,490,826	34.32 / 34.73	10.28 / 10.36	21.07 / 21.27

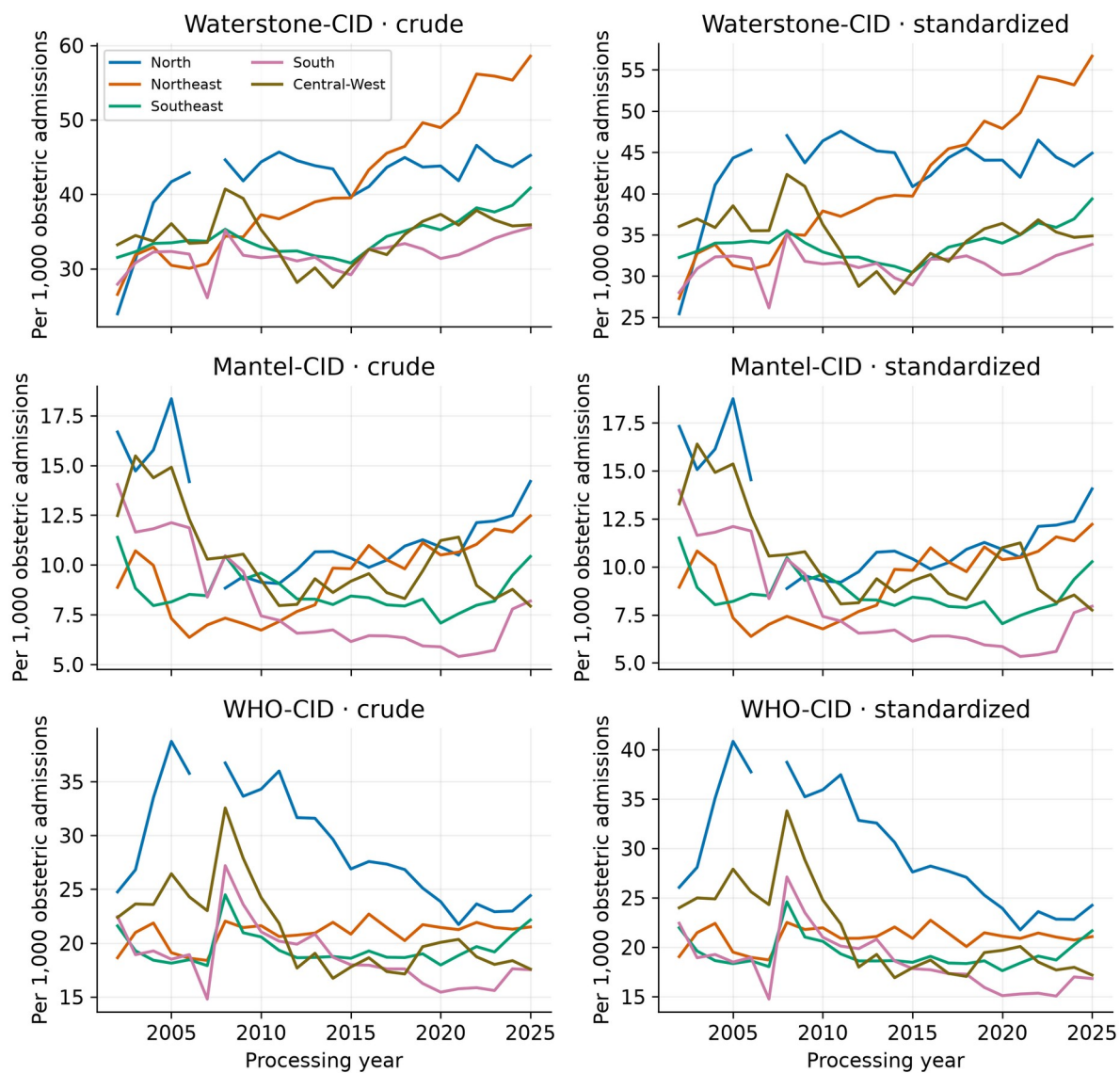


Figure 1. Regional frequencies before and after direct age standardization. Annual curves require 12 complete processing months; the North has a gap in 2007. All panels use linear scales, with panel-specific ranges. Standardization uses eight five-year age groups and the same national weights in every year.

Age, diagnostic composition and list overlap

All three national age profiles were higher at ages 10–14 than at 15–19, and rose across the older adult groups. At ages 45–49, frequencies reached 73.07 per 1,000 for Waterstone-CID, 24.85 for Mantel-CID and 51.98 for WHO-CID. The share of all eligible admissions at ages 35–49 increased from 8.40% in 2002 to 15.08% in 2025. Table 2 gives the descriptive comparison of admissions at ages 35–49 and 10–34 years. Figure 2 displays the full age profiles. Pre-eclampsia/superimposed-hypertension codes accounted for 1,038,088 (48.19%) of Waterstone-

CID identifications; adding eclampsia brought the hypertensive group to 1,198,073 (55.62%). Hemorrhage-related codes accounted for 498,679 (23.15%) and infection-related codes for 437,073 (20.29%).

Table 2. Identifications by age and frequency ratios, 2002–2025. Numerators count surviving admissions with the marker; denominators include all eligible obstetric admissions in each age category. CIs are conventional and do not account for repeat admissions of the same woman.

List	10–34 years, n/N	35–49 years, n/N	Frequency ratio (95% CI)
Waterstone-CID	1,793,839/51,891,828	360,208/6,485,089	1.61 (1.60–1.61)
Mantel-CID	464,435/51,891,828	77,719/6,485,089	1.34 (1.33–1.35)
WHO-CID	1,041,312/51,891,828	189,998/6,485,089	1.46 (1.45–1.47)

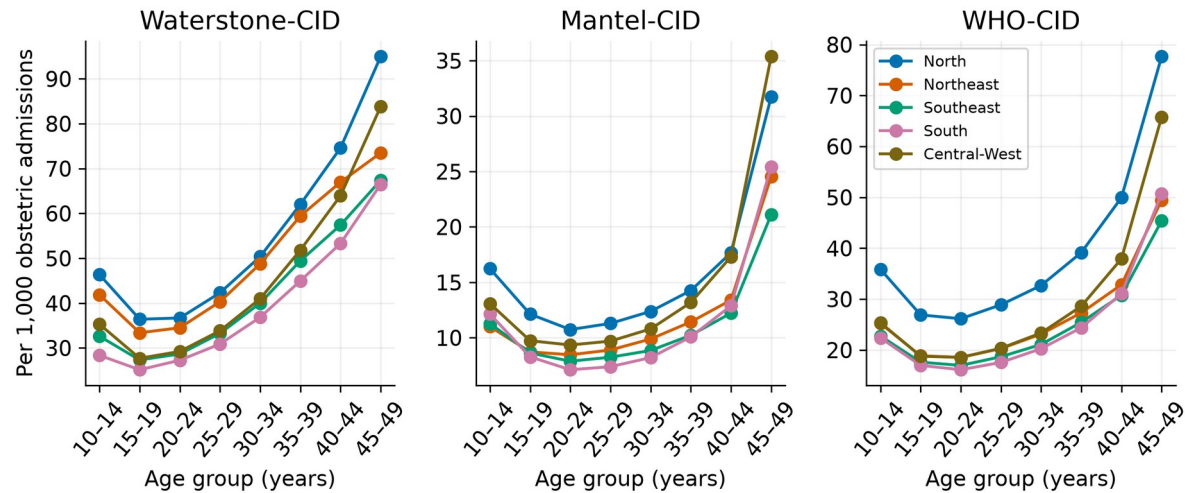


Figure 2. Age-specific frequencies by region and diagnostic list. Pooled estimates use the 287 common processing months. The horizontal axis represents five-year age groups; frequencies are specific to each group and are not standardized.

Recorded ICU use among identified admissions with known ICU status was 4.08% for Waterstone-CID; 1.74% for Mantel-CID; 2.85% for WHO-CID. The full denominators and unknown counts are supplied in the aggregate tables. The three-way intersection comprised 455,700 admissions. Waterstone-CID alone accounted for 1,039,809, whereas almost all Mantel-CID identifications also met WHO-CID: only 30 were outside that list. Jaccard similarity was 0.203 for Waterstone-CID/Mantel-CID, 0.491 for Waterstone-CID/WHO-CID

and 0.440 for Mantel-CID/WHO-CID (Figure 3).

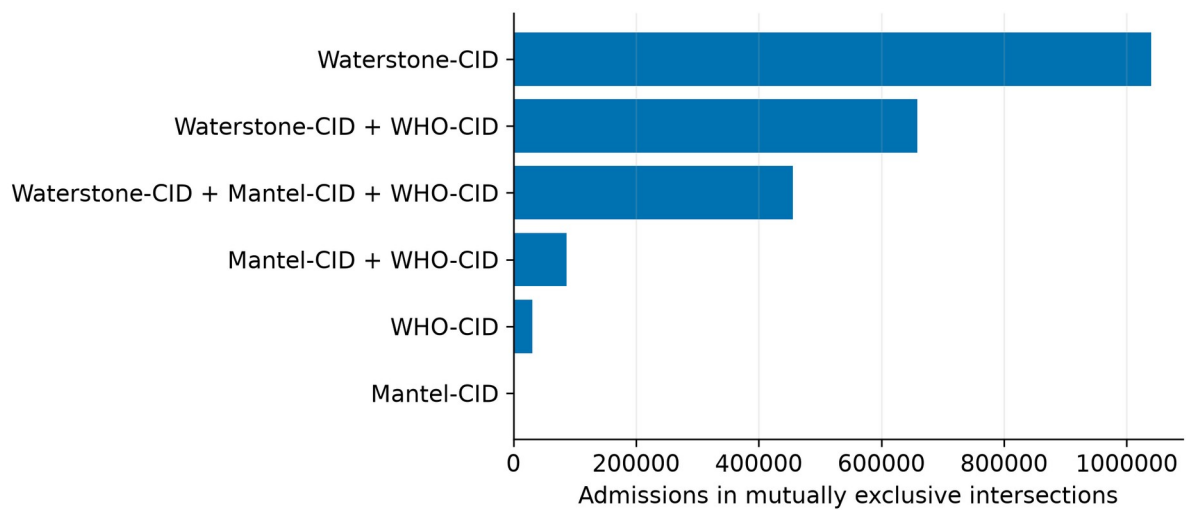


Figure 3. Mutually exclusive positive combinations of the three diagnostic lists. Each admission appears in one bar. The bars together represent the union of positive identifications, not the sum of list-specific totals.

Sensitivity to population and diagnosis fields

Among admissions with non-zero prenatal registration, frequencies were 17.44 per 1,000 for Waterstone-CID, 0.99 per 1,000 for Mantel-CID, 6.12 per 1,000 for WHO-CID. Prenatal-field completeness varied from 9.82% in the Northeast to 29.58% in the South. Within the registered subgroup, the Northeast had the highest Waterstone-CID frequency, while the South had the highest Mantel-CID and WHO-CID frequencies. This contrasted with the North's highest pooled frequencies in the full obstetric population (Figure 4). Using any available diagnosis instead of the principal diagnosis yielded 39.31 per 1,000 for Waterstone-CID, 9.88 per 1,000 for Mantel-CID, 22.42 per 1,000 for WHO-CID. Restricting denominators to surviving admissions yielded 36.91 per 1,000, 9.29 per 1,000, 21.10 per 1,000, respectively.

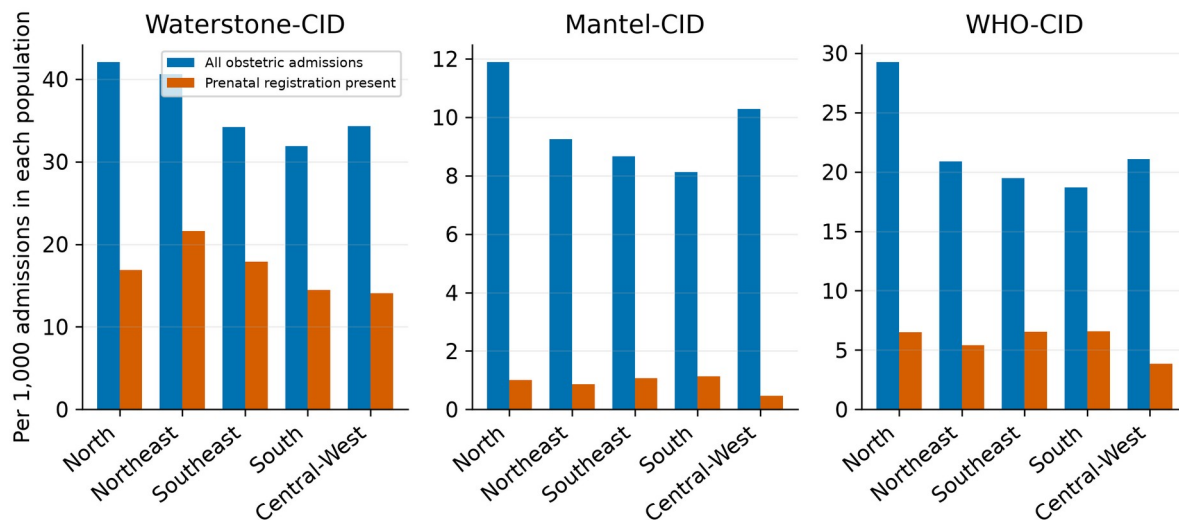


Figure 4. Frequencies in the full obstetric population and the subgroup with a recorded prenatal-registration number. Numerators and denominators use the same subgroup, region and 287 common months. Both sets are crude frequencies. Registration is not a measure of prenatal-care attendance.

Regional socioeconomic associations

Of 90 crude-frequency Pearson comparisons, 19 had BH-adjusted $q < 0.05$. The separate age-standardized sensitivity family had 25 such comparisons. Pairwise sample sizes ranged from 5 to 14 years. Among the 19 crude-frequency findings with $q < 0.05$, seven reversed sign after first differencing. For example, the Northeast correlation between nominal PNAD income and Waterstone-CID was $r = 0.932$ in levels and $r = 0.047$ in consecutive differences. Some associations remained appreciable: PNAD Contínua real income and Mantel-CID had correlations of 0.911 in levels and 0.791 in differences in the South, compared with -0.684 and -0.619 in the Central-West. Thus, neither direction nor sensitivity to temporal transformation was uniform across regions. All coefficients, confidence intervals, multiplicity-adjusted values, years and paired observations are available in the supplementary tables.

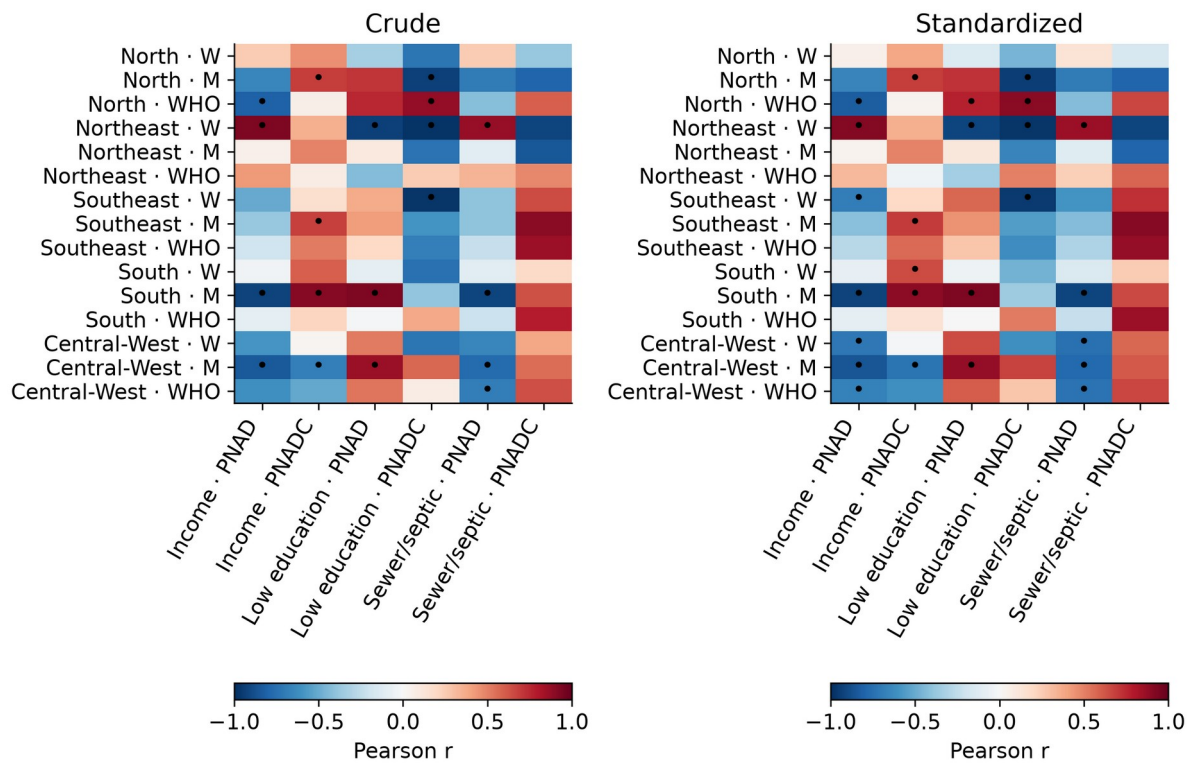


Figure 5. Pearson correlations between regional socioeconomic indicators and administrative frequencies. Left: crude frequencies; right: age-standardized frequencies. Each panel is a separate family of 90 comparisons. Dots indicate BH-adjusted $q < 0.05$. PNAD and PNAD Contínua (PNADC) series have different definitions and are not concatenated. Conventional inference does not account for temporal autocorrelation.

Municipal analysis in Rio Grande do Norte

The RN complement included 221,297 admissions across 167 municipalities in 2008–2012. The state-level frequency of the Rosendo marker set was 46.64 per 1,000 (10,321 identifications); hypertensive and hemorrhagic frequencies were 34.19 and 6.94 per 1,000. The unweighted municipal mean for the overall marker set was 38.07 per 1,000. None of the nine municipal Pearson comparisons had BH-adjusted $q < 0.05$. Overall marker frequency was essentially uncorrelated with income per capita ($r = 0.004$). Restriction to the 36 municipalities with at least 1,000 admissions yielded no Pearson comparison with $q < 0.05$. A negative rank correlation between income and hypertensive-marker frequency appeared in the full municipal sample (Spearman $\rho = -0.231$; unadjusted $p = 0.0027$), illustrating sensitivity to the correlation measure; this exploratory rank result was not included in the multiplicity-adjusted Pearson family. Supplementary Figure S2 displays the municipal distributions.

Discussion

Across more than 58 million obstetric admissions, administrative near-miss markers showed substantial variation by diagnostic list, region and age. The North had the highest pooled frequencies under all three definitions, and the South the lowest, after standardization as well as before it. The national Waterstone-CID frequency was higher in 2025 than in 2002 even after accounting for age composition, whereas Mantel-CID and WHO-CID had similar endpoint magnitudes. Hypertensive diagnoses accounted for more than half of Waterstone-CID identifications. These patterns identify dimensions of maternal morbidity visible in public hospital data without equating diagnosis codes with clinically confirmed near miss.

Prenatal-registration selection was especially consequential. Requiring a non-zero registration number retained only about one sixth of eligible obstetric admissions and changed both marker frequencies and regional rankings. Because completeness differed between regions, this restriction described different selected populations across the country. The contrast is evidence about case ascertainment and administrative selection; it cannot establish a protective effect of prenatal care or distinguish nonattendance from an absent registration number.

Relationship to Brazilian and international evidence

Administrative studies provide the closest comparison for the present results. Carvalho and colleagues analyzed national Waterstone-based patterns in 2000–2012, while Herdt and colleagues studied regional variation in SIH admissions in 2010–2018 [5, 6]. Differences in eligibility, diagnosis fields, procedures, age limits and denominators prevent direct interpretation of differences in absolute frequencies as changes in clinical disease burden. The useful comparison concerns the pattern of recorded conditions and inequalities within clearly specified administrative populations.

The RN complement preserved the original diagnostic approach, age limits and study period of Rosendo and Roncalli [4], but it addressed municipalities directly. Its statewide frequency of 46.64 per 1,000 and unweighted municipal mean of 38.07 should not be compared as equivalent estimates with the original mean of 36.67. Rosendo and Roncalli formed spatial clusters, used population-weighted municipal frequencies within those clusters and excluded two outlying clusters, analyzing 63 units. Our 167-municipality analysis used admission-weighted pooling for the statewide estimate and no outcome-based exclusions. Different geographic aggregation, covariate definitions and weighting can therefore explain differences without requiring a change

in the underlying phenomenon.

The absence of a multiplicity-adjusted municipal Pearson finding did not reproduce a simple income gradient for the overall marker set. It does not refute the original contextual analysis: we did not reconstruct its spatial clusters, complete socioeconomic and care-coverage variables, or multivariable models. The negative rank association between income and hypertensive markers warrants cautious attention, but conventional municipal tests do not address spatial dependence. This complement is useful for showing how conclusions depend on scale, outcome and method, rather than serving as an exact replication or validation of the earlier study.

Clinical evidence answers a different but essential question. The Birth in Brazil survey associated maternal near miss with individual and care-related factors [14], while hospital studies show that the performance of administrative identification depends on the information available and the clinical reference used [7, 8]. Our analysis does not estimate sensitivity or specificity against a clinical standard. Nor does agreement between two overlapping lists establish accuracy: common codes can produce high overlap even when both definitions lack the clinical information needed to classify severity.

International evidence likewise emphasizes definition and context. Firoz and colleagues documented substantial variation in the application and adaptation of near-miss criteria across settings [15]. The prospective Latin American study of organ-dysfunction definitions by Rojas-Suarez and colleagues used clinical observations that are not available in reduced SIH records [16]. These studies support transparent administrative surveillance alongside clinical investigation; their populations and denominators do not provide a universal numerical benchmark for the present frequencies. Recent Brazilian work on state surveillance [17], and proposals for national surveillance and clinical data collection [18, 19], identify possible routes toward linking these complementary forms of evidence.

Interpretation and limitations

The increased proportion of admissions at older ages contributed to differences between crude and standardized frequencies, but age composition did not account for the entire Waterstone-CID endpoint increase or the pooled regional ordering. The higher frequencies among both the youngest adolescents and older women merit more focused study; they concern hospitalized populations and cannot be translated directly into pregnancy-level risks. The very narrow conventional confidence intervals for age contrasts reflect the large number of records, not the

absence of measurement error or selection.

Differences between diagnostic lists were structural as well as numerical. The near-complete inclusion of Mantel-CID identifications within WHO-CID shows why these outputs should not be interpreted as independent confirmations of severity. Similarly, principal-diagnosis frequencies were lower than frequencies based on any recorded diagnosis, but the latter also depend on the availability and completeness of secondary fields. We retained the principal-diagnosis definition for the main comparisons and reported the broader definition as a sensitivity analysis.

Socioeconomic correlations must be interpreted at their measured scale. A regional correlation between annual averages cannot identify an individual woman's exposure or risk. Secular changes in income, education, sanitation, access and coding can generate correlations even without a direct relationship between the two variables studied. Separate survey definitions, conventional uncertainty estimates and few annual observations further limit inference. First differences probe sensitivity to shared trends but do not remove all time-varying confounding or autocorrelation. Municipal comparisons additionally depend on geographic aggregation and neighboring areas may not be statistically independent.

The age-standardized analysis addresses one component of changing admission composition. It does not adjust for parity, comorbidity, referral severity, private-sector use, availability of intensive care or other determinants. Hospital-based geographic frequencies can be affected by cross-border referral. Recorded ICU use reflects both illness and access to beds and does not itself establish organ dysfunction.

The denominator is hospital admissions financed by SUS, so changes can reflect the composition and volume of obstetric hospital care rather than the risk among all pregnancies. Repeat admissions of the same woman remain possible. Processing dates and later submissions may shift the apparent timing of events; complete calendar coverage does not guarantee equal administrative maturity across years. The absence of one historical state-month was handled explicitly, but unmeasured underrecording within available files remains possible. The pandemic may also have altered case mix, service use and coding; a national administrative curve cannot isolate its causal effect, and clinically selected COVID-19 cohorts are not directly comparable [20].

Finally, the analysis ends at discharge and cannot describe recovery, disability or later deaths.

Studies of post-near-miss follow-up emphasize outcomes beyond survival [21]. Administrative monitoring is therefore most useful for identifying patterns that warrant focused investigation, supporting transparent surveillance and directing questions to clinical and population-based studies.

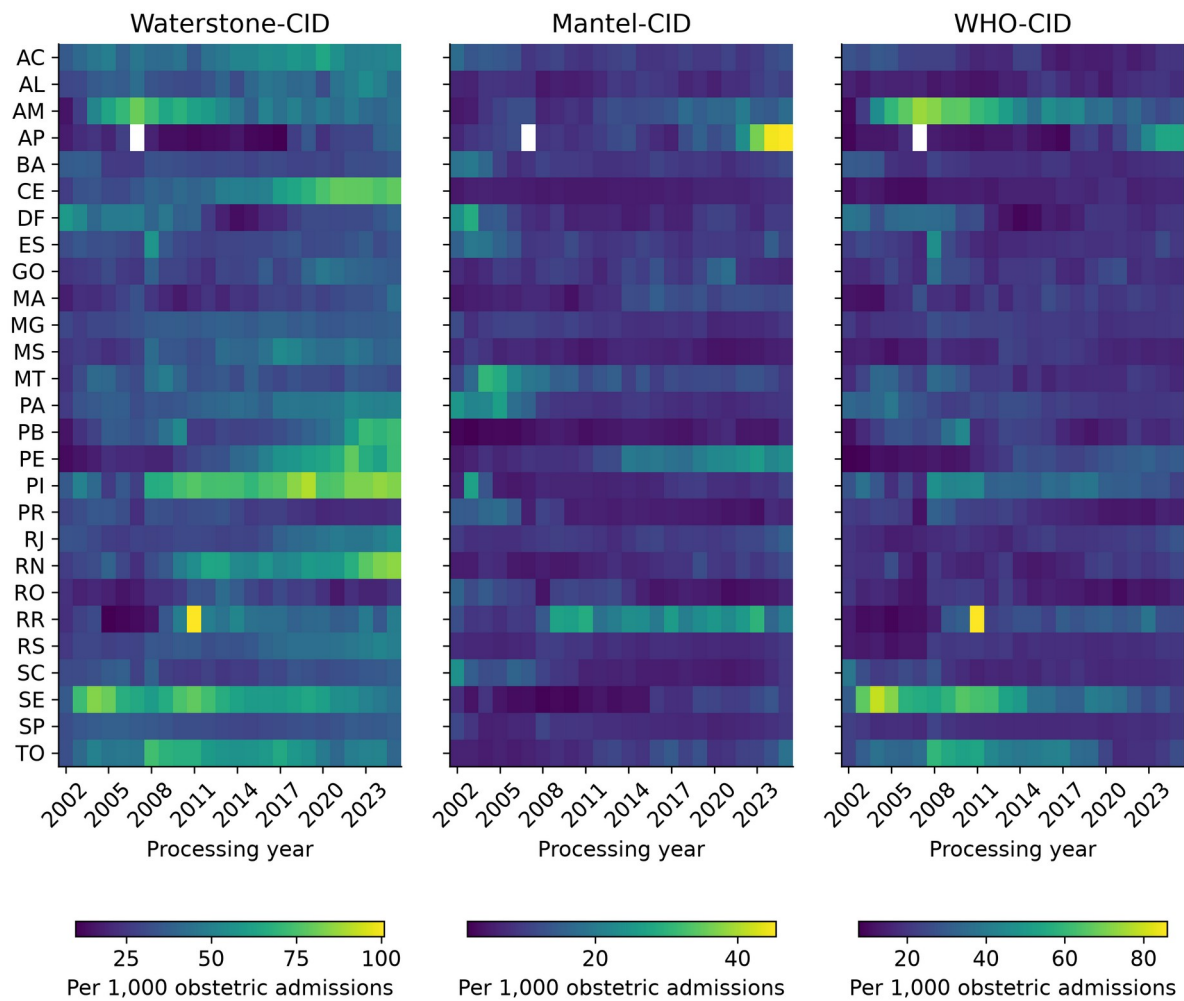
Conclusion

SIH-SUS data from 2002–2025 identify regional and age variation in administrative markers of maternal near miss, with hypertensive diagnoses contributing prominently to the Waterstone-derived list. Age standardization preserved the higher pooled frequencies in the North, while requiring prenatal registration substantially changed the population and its regional comparisons. Socioeconomic associations depended on region, definition, temporal transformation and geographic scale. Clearly specified admission populations, diagnostic lists and denominators make these data useful for surveillance and hypothesis generation; explanations of clinical severity and contextual mechanisms require complementary evidence.

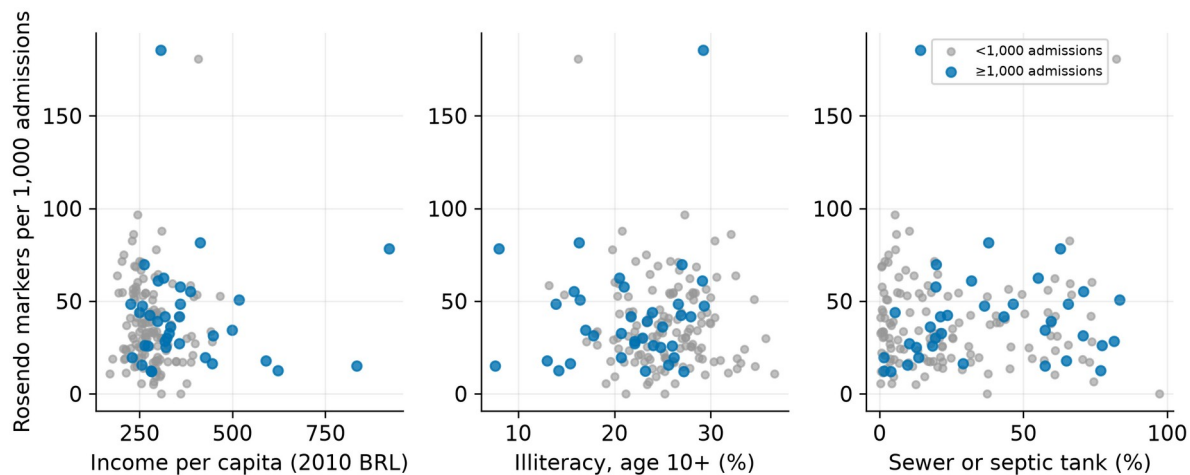
Data and analytical materials

The analytical materials in the [working repository](#) contain the codebook, source inventory, coverage matrix, age-standardization weights, all coefficient estimates and paired observations, and supplementary tables. The code operates on public DATASUS and IBGE data. Counts refer to admission authorizations; no person-level data are included in the dissemination dashboard.

Supplementary figures



Supplementary Figure S1. State-level crude frequencies by processing year. Each panel has its own color scale. White cells indicate incomplete annual coverage, not zero frequency. AP/2007 is incomplete. These descriptive patterns do not adjust for referral flows or differences in recording.



Supplementary Figure S2. Municipal Rosendo-list frequencies and 2010 Census indicators, RN, 2008–2012. Each point is one municipality. Blue points have at least 1,000 eligible admissions; gray points have fewer. The three panels use the same overall outcome and do not represent the original study's spatial clusters. No fitted regression line or causal effect is implied.

References

1. Say L, Souza JP, Pattinson RC, et al. Maternal near miss--towards a standard tool for monitoring quality of maternal health care. *Best Pract Res Clin Obstet Gynaecol.* 2009;23(3):287-96. [doi:10.1016/j.bpobgyn.2009.01.007](https://doi.org/10.1016/j.bpobgyn.2009.01.007).
2. Mantel GD, Buchmann E, Rees H, et al. Severe acute maternal morbidity: a pilot study of a definition for a near-miss. *Br J Obstet Gynaecol.* 1998;105(9):985-90. [doi:10.1111/j.1471-0528.1998.tb10262.x](https://doi.org/10.1111/j.1471-0528.1998.tb10262.x).
3. Waterstone M, Bewley S, Wolfe C. Incidence and predictors of severe obstetric morbidity: case-control study. *BMJ.* 2001;322(7294):1089-93; discussion 1093-4. [doi:10.1136/bmj.322.7294.1089](https://doi.org/10.1136/bmj.322.7294.1089).
4. Rosendo TM, Roncalli AG. [Maternal near misses and health inequalities: an analysis of contextual determinants in the State of Rio Grande do Norte, Brazil]. *Cien Saude Colet.* 2016;21(1):191-201. [doi:10.1590/1413-81232015211.20802014](https://doi.org/10.1590/1413-81232015211.20802014).
5. Carvalho BAS, Andrade AGBF, Dantas AS, et al. Temporal trends of maternal near miss in Brazil between 2000 and 2012. *Rev Bras Saude Mater Infant.* 2019;19:115–124. [doi:10.1590/1806-93042019000100007](https://doi.org/10.1590/1806-93042019000100007).

6. Herdt MCW, Magajewski FRL, Linzmeyer A, et al. Temporal Trend of Near Miss and its Regional Variations in Brazil from 2010 to 2018. *Rev Bras Ginecol Obstet*. 2021;43(2):97-106. [doi:10.1055/s-0040-1719144](https://doi.org/10.1055/s-0040-1719144).
7. Nakamura-Pereira M, Mendes-Silva W, Dias MA, et al. [The Hospital Information System of the Brazilian Unified National Health System: a performance evaluation for auditing maternal near miss]. *Cad Saude Publica*. 2013;29(7):1333-45. [doi:10.1590/s0102-311x2013000700008](https://doi.org/10.1590/s0102-311x2013000700008).
8. Lobato G, Nakamura-Pereira M, Mendes-Silva W, et al. Comparing different diagnostic approaches to severe maternal morbidity and near-miss: a pilot study in a Brazilian tertiary hospital. *Eur J Obstet Gynecol Reprod Biol*. 2013;167(1):24-8. [doi:10.1016/j.ejogrb.2012.10.030](https://doi.org/10.1016/j.ejogrb.2012.10.030).
9. Rosendo TS, Roncalli AG, Azevedo GD. Prevalence of Maternal Morbidity and Its Association with Socioeconomic Factors: A Population-based Survey of a City in Northeastern Brazil. *Rev Bras Ginecol Obstet*. 2017;39(11):587-595. [doi:10.1055/s-0037-1606246](https://doi.org/10.1055/s-0037-1606246).
10. Ministério da Saúde, DATASUS. SIH-SUS reduced hospital files, field dictionary IT_SIHSUS_1603 and TAB_SIH conversion tables. Snapshot: 16 September 2026. [Official data-distribution portal](#).
11. Instituto Brasileiro de Geografia e Estatística. Aggregate-data API, version 3; SIDRA tables 1860, 272, 1956, 7437, 7133, 7192, 3578, 1383 and 1394. Regional snapshot: 16 September 2026; municipal snapshot: 18 September 2026. [API documentation](#).
12. Instituto Brasileiro de Geografia e Estatística. PNAD 2004: ocupação cresceu e rendimento ficou estável. 2005. [Geographic coverage expansion](#).
13. Sousa MH, Cecatti JG, Hardy EE, Serruya SJ. Severe maternal morbidity (near miss) as a sentinel event of maternal death. An attempt to use routine data for surveillance. *Reprod Health*. 2008;5:6. [doi:10.1186/1742-4755-5-6](https://doi.org/10.1186/1742-4755-5-6).
14. Domingues RM, Dias MA, Schilithz AO, et al. Factors associated with maternal near miss in childbirth and the postpartum period: findings from the birth in Brazil

National Survey, 2011-2012. *Reprod Health*. 2016;13(Suppl 3):115.

[doi:10.1186/s12978-016-0232-y](https://doi.org/10.1186/s12978-016-0232-y).

15. Firoz T, Trigo Romero CL, Leung C, et al. Global and regional estimates of maternal near miss: a systematic review, meta-analysis and experiences with application. *BMJ Glob Health*. 2022;7(4):e007077. [doi:10.1136/bmjgh-2021-007077](https://doi.org/10.1136/bmjgh-2021-007077).
16. Rojas-Suarez J, Santacruz J, Pajaro Y, et al. Development of a new definition of maternal near miss based on organ dysfunction in Latin America and the Caribbean: A prospective multicenter cohort study. *Int J Gynaecol Obstet*. 2025;168(1):155-166. [doi:10.1002/ijgo.15818](https://doi.org/10.1002/ijgo.15818).
17. Crestani APJ, Welter Wendt G, Mazur CE, et al. Surviving Severe Obstetric Complications: A Population-Based Analysis of Maternal near Miss. *Med Sci (Basel)*. 2026;14(2):313. [doi:10.3390/medsci14020313](https://doi.org/10.3390/medsci14020313).
18. Ferreira MES, Coutinho RZ, Queiroz BL. [Maternal morbidity and mortality in Brazil and the urgency of a national surveillance system for maternal near miss]. *Cad Saude Publica*. 2023;39(8):e00013923. [doi:10.1590/0102-311XPT013923](https://doi.org/10.1590/0102-311XPT013923).
19. Domingues RMSM, Dias MAB, Nakamura-Pereira M, et al. Perinatal mortality, severe maternal morbidity and maternal near miss: protocol of a study integrated with the Birth in Brazil II survey. *Cad Saude Publica*. 2024;40(4):e00248222. [doi:10.1590/0102-311XPT248222](https://doi.org/10.1590/0102-311XPT248222).
20. Cunha CMP, Amorim MMR, Guendler JA, et al. Clinical and epidemiological profile of pregnant and postpartum women affected by COVID-19 who required respiratory support. *Rev Bras Ginecol Obstet*. 2025;47. [doi:10.61622/rbgo/2025rbgo14](https://doi.org/10.61622/rbgo/2025rbgo14).
21. Singla N, Patil AN, Mehra A, et al. Hospital-Based Multidisciplinary Postnatal Care for Holistic Maternal Recovery: A Collaborative Assessment of Long-Term Outcomes after Maternal Near Miss. *Hosp Top*. 2026. Online publication: 1–9. [doi:10.1080/00185868.2026.2724240](https://doi.org/10.1080/00185868.2026.2724240).